

RelayDrop

“Low-cost, off-grid communication network for small fishing boats beyond reliable cellular coverage”

Shore-based Command Center

Relay Node

Flagship Node

Warship Node

Airborne Node

Buoy – shore Link

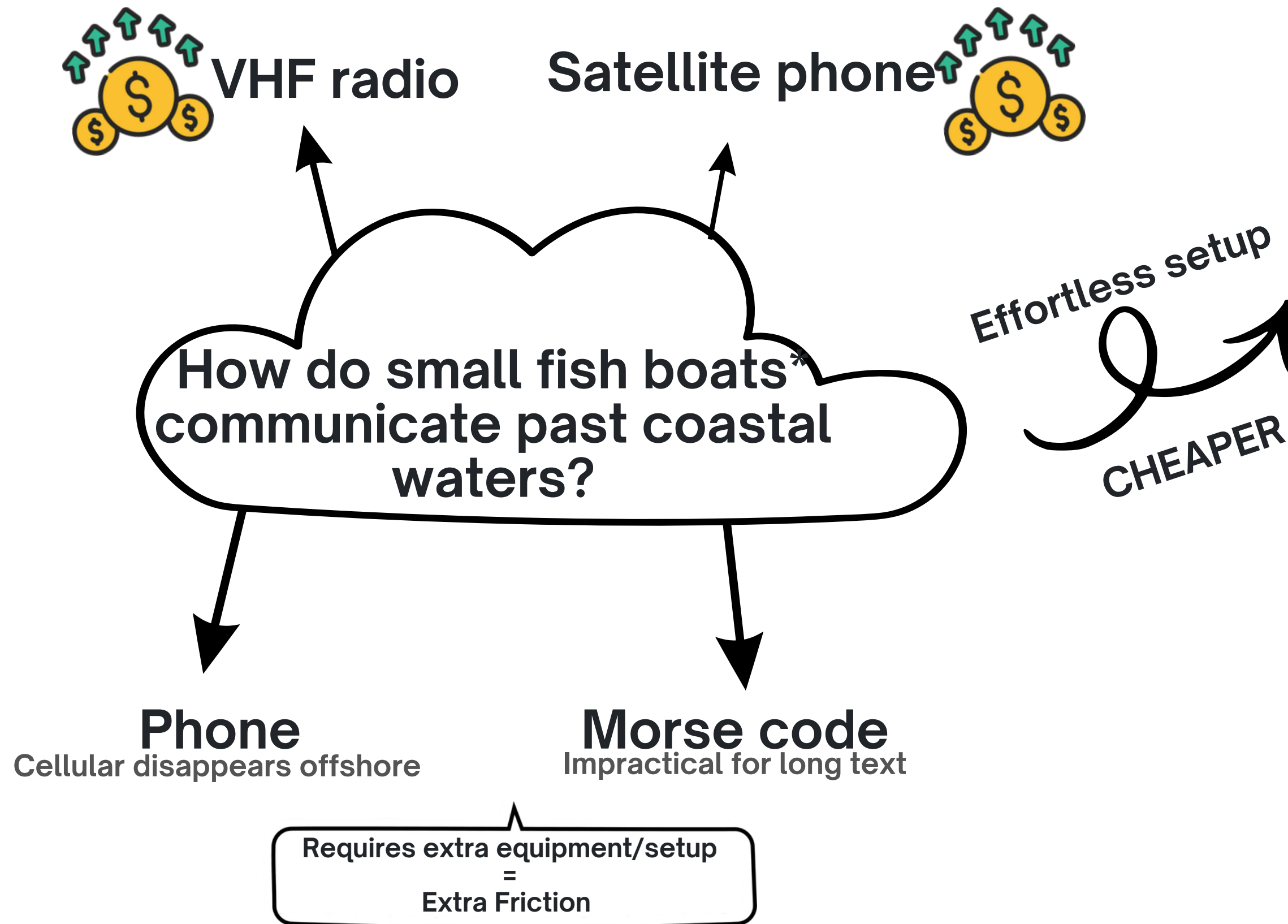
Buoy – Vessel Link

Vessel – shore Link

Team Details*	Name	Faculty and Department	Student Number	Year of Study in AY2026/2027	Role in Team	Key Contributions
Team Leader	Manan Mishra	CDE	A0333037M	Year 2	RF & Systems Engineer	Led engineering design and technical integration
Member 1	Yash Mittal	SOC	A0360161R	Year 1	Software Developer	Developed firmware, integrated website, and performed link & stress testing
Member 2	Atharva Shukla	SOC+CDE	A0336919N	Year 2	Technical & Strategy Lead	Business planning, outreach, critical review, and technical validation
Member 3	Pranit Macker	BIZ+FOS	A0334213U	Year 2	Business & Marketing Lead	Business plan, marketing strategy, and market analysis

* Minimum of 1 CDE student is required per EUREKA team

The Problem



Introducing RelayDrop



Simple for crew
+
Affordable for fleet
+
Works without cellular

*90% of the global capture fisheries workforce

The cooperative installs the hardware once. Crew simply connect and send.

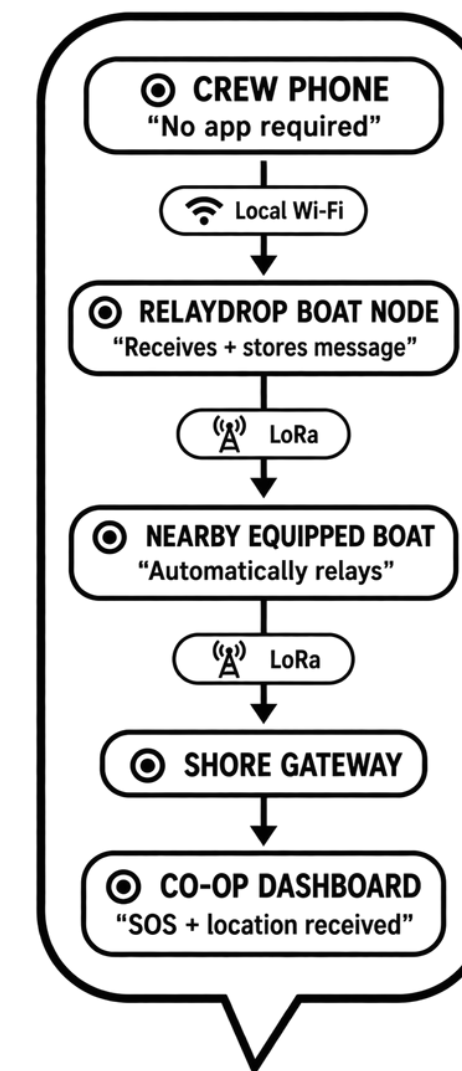


Internet

Works without cellular
→ LoRa boat-to-boat relay

Affordable for fleet
→ shared cooperative infrastructure

How it works?

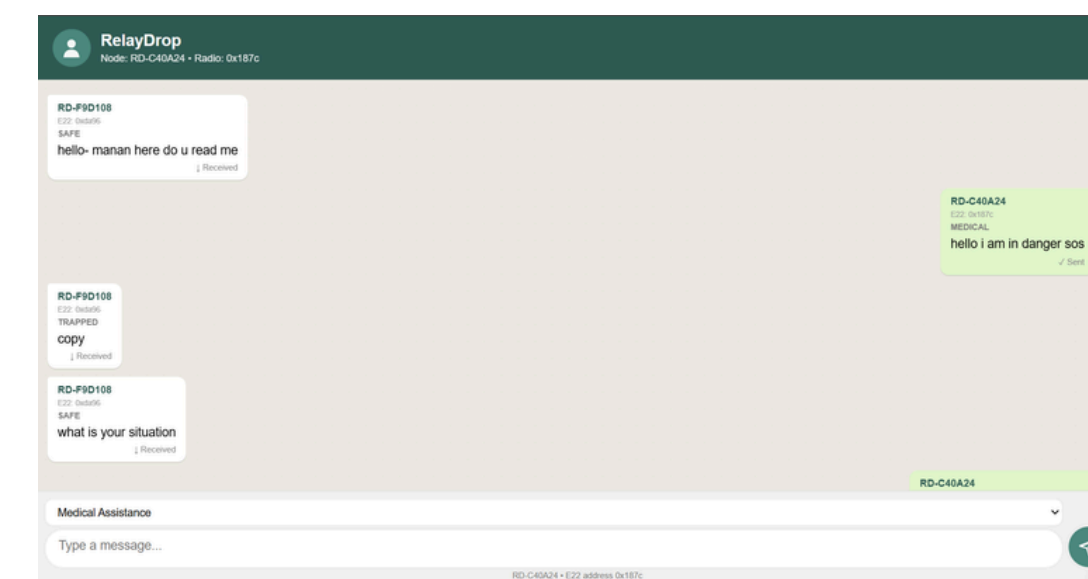


900 Mhz



Simple for crew

→ No app. No account. No radio training.

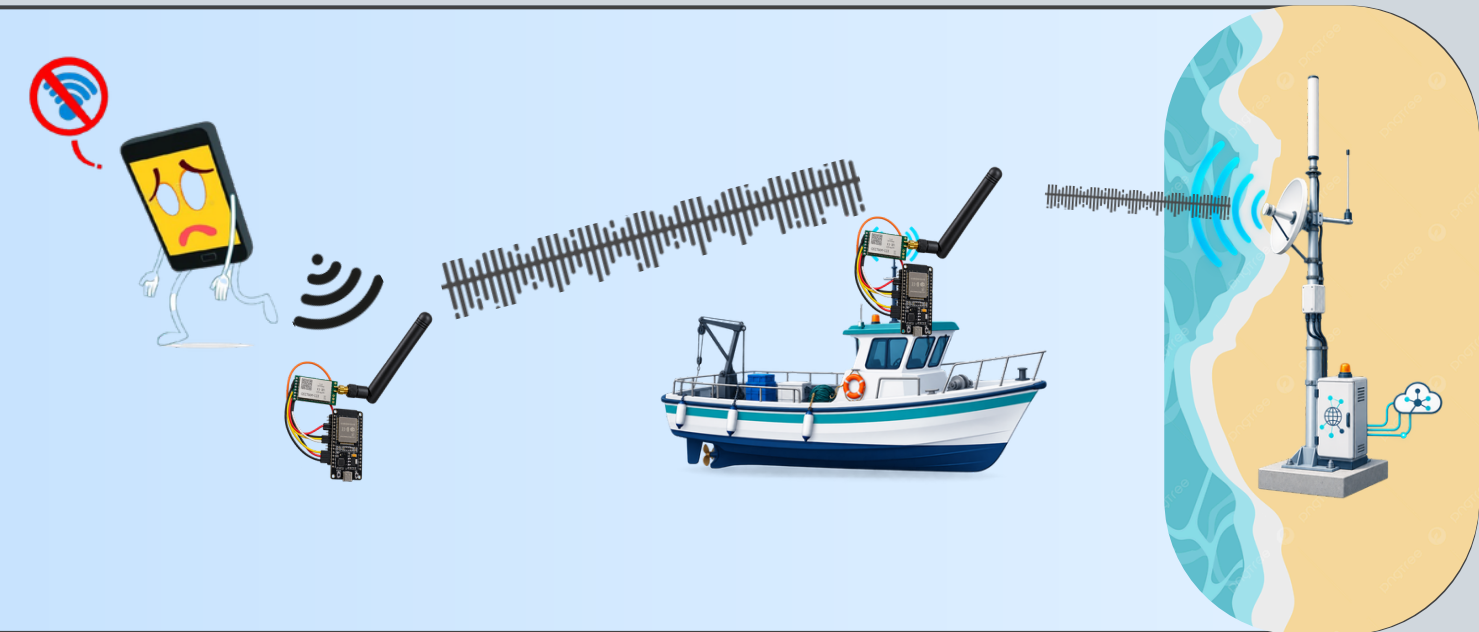


No internet required

01

Innovation

- RelayDrop turns the phone a fisher already owns into a satellite-free emergency terminal.
- The crew opens a webpage (through the intranet bubble) → submits a message → Neighbouring boats/buoys relay it shoreward → reaches a coastal gateway.
- The innovation is not the radio. **It is making off-grid maritime alerting work at a price point and usability level that a 9-metre boat can actually adopt.**



02

Only We Offer This

- Starlink and Apple direct-to-cell need a local carrier and a paid subscription, which is unavailable across most of Indonesian and Philippine coastal telco coverage.
- VHF and AIS carry licensing, cost, and training burdens, and international safety conventions only bind vessels 24 metres and above.
- Mesh radios like Meshtastic and goTenna require users to configure a device in advance.
- RelayDrop is the only option that needs **zero additional subscription, zero app installation, and zero technical literacy from the end user.**



Meshtastic-style personal mesh

User setup required

- 1 Buy device
- 2 Configure device
- 3 Install app
- 4 Pair / learn
- 5 Communicate

VS



RelayDrop

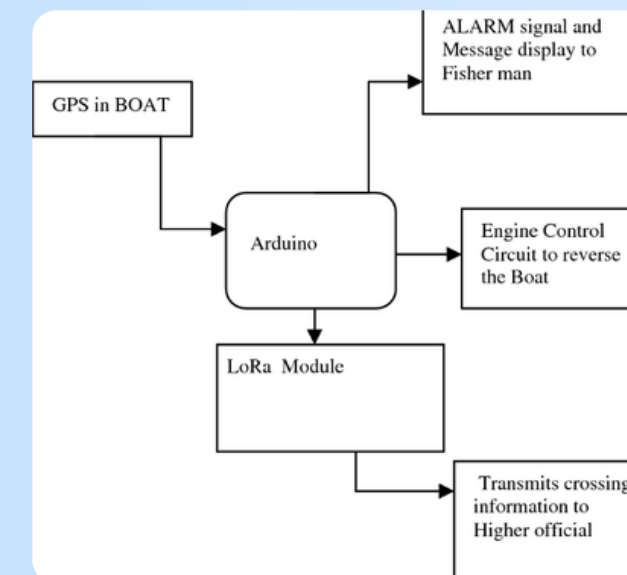
Simple for the crew

- 1 Co-op installs once
- 2 Crew joins Wi-Fi
- 3 Sends

03

Our Secret

- Most maritime LoRa projects like C-Mesh in Portugal, the Azores network, etc. are strong research prototypes **that never shipped**. We identified that the binding constraint is **user adoption friction** and designed around this from the start.
- Singapore places us inside one of the **world's densest maritime clusters**: MPA, PIER71, Singapore Maritime Institute, and the beachhead fleets in Indonesia and the Philippines are a short flight away for field validation.



Size of Market

USER



Crews on vessels under ~12m operating 10–40 km offshore, typically with no AIS, no satcom, and no licensed VHF

BUYER



Fishing co-operatives, marine insurers, NGOs and development programmes working on fisher safety

1.1M

fishing vessels in Indonesia alone

9,394

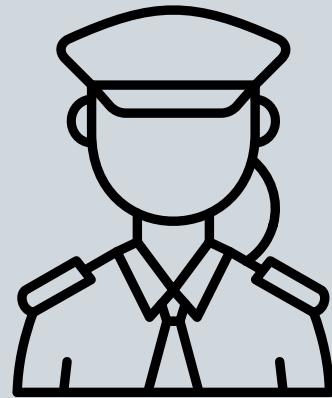
Indonesian vessels fitted with VMS
(Jan 2026)

Proof of demand is here.

- Market scale: No. of target vessels in each province × expected unit price × realistic adoption rate. Then expand the model province by province.
- Market trend: Indonesia is already expanding Vessel Monitoring System (VMS) adoption, driven by safety, operations, and falling hardware costs. The Demand is here.
- Market future: Proof of demand already exists. RelayDrop extends this safety-tech market to smaller vessels priced out of conventional VMS and satellite systems.

- Philippines adds 407,215 registered municipal vessels (BFAR, 2023). Combined beachhead of ~1.5M vessels across Indonesia + Philippines.
- Globally, small-scale fisheries support 1 in every 12 people and generate \$77.2B in landed economic value (Nature, 2024).

Market Scale: Magnified



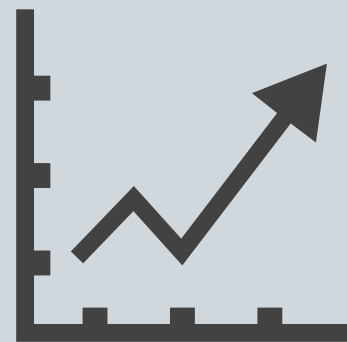
Simulate a Pilot

Pilot first, then sell through the co-operative.
Run a subsidised pilot with 20–30 vessels and one coastal gateway, measuring packet delivery, alert latency, reliability, and real-world incidents to build a documented case study. Also hope to approach marine insurers through this study.



Co - op trust

After validation, the co-operative becomes the customer. Rather than selling vessel-by-vessel, RelayDrop offers the co-op a paid fleet package with premium services. The co-op already has the trust, so one institutional sale can deploy RelayDrop across dozens of boats.

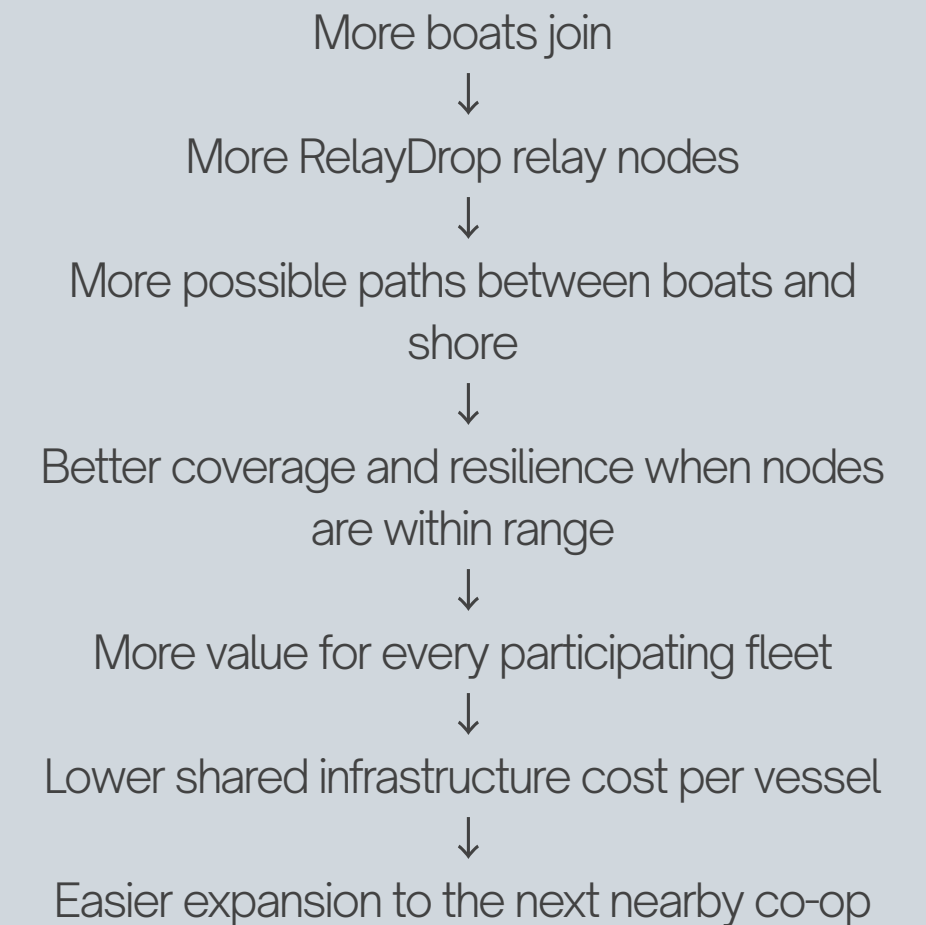


Low-Cost Hardware, Shared Infrastructure

Current prototype core BOM: ~**SGD 19.32/unit**
Sell for under SGD \$50; co-ops pay per vessel for monitoring, alerts and the dashboard. Shared shore gateways spread infrastructure cost across the whole fleet.

Scale by Densifying the Network

The network gets stronger as adoption grows:



Growth creates its own infrastructure: more vessels create more relay paths, denser coverage and lower shared cost per vessel.

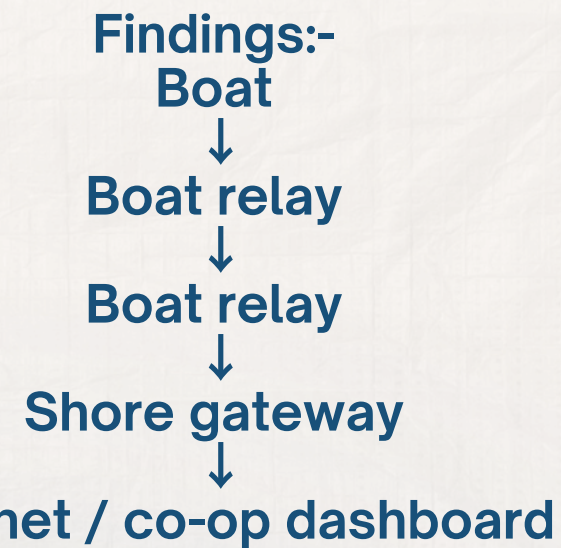
WHY EXISTING SOLUTIONS LEAVE A GAP

Most marine connectivity is sold vessel-by-vessel. RelayDrop is deployed fleet-by-fleet.

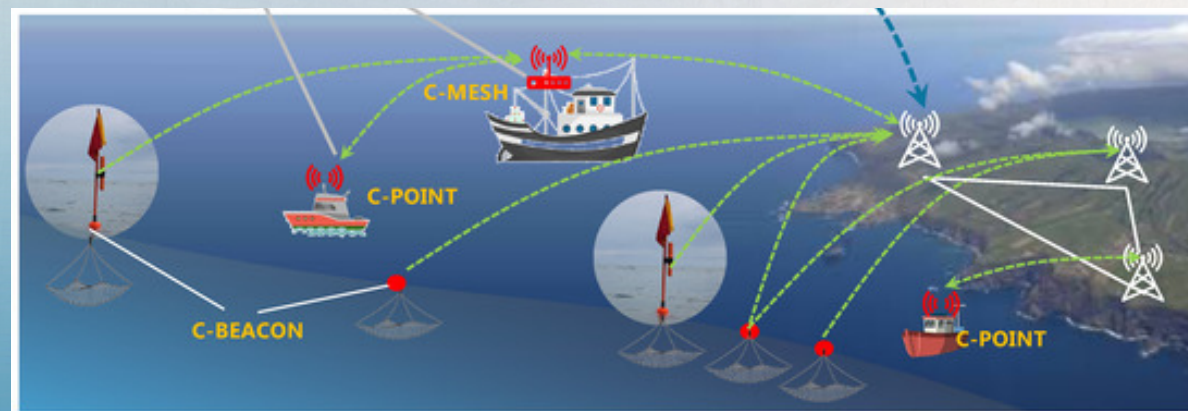
Parameters	Traditional marine systems (Satellite&radios)	Maritime LoRa research projects (e-Nalayan, C-Mesh)	Private Companies (Mechtastic, goTenna Pro)	RelayDrop
Definition	Dedicated vessel communication or tracking systems using radio, AIS, or satellite links, Expensive	Research and pilot networks using LoRa to connect vessels and shore over long distances	Personal open source radio devices that let users exchange messages without cellular coverage	Low-cost fleet network where boats relay messages between each other and shared shore gateways
Designed around	Individual vessel equipment	Whole fishing fleet	Individual users	Whole fishing fleet
Deployment unit	Vessel by vessel	Research / pilot deployments	Person by person	Co-op / fleet deployment
User interaction	Dedicated marine equipment- High friction setup	Fisher uses existing phone	Dedicated device/app setup (needs internet)	Fisher uses existing phone
Network behaviour	Each vessel mainly buys its own capability	Each participating vessel also strengthens fleet coverage	Users form an ad-hoc mesh	Each participating vessel also strengthens fleet coverage
Shore visibility	Depends on system/service	Built around boat ↔ fleet ↔ shore connectivity	Usually secondary	Built around boat ↔ fleet ↔ shore connectivity
Commercial model	Sell equipment/service to vessel	Research-led; limited commercialisation	Sell device to consumer	Sell fleet package + shared infrastructure to co-op
Vessel owner / operator	Research body / public programme	Research organisation	Individual user	Co-op / fleet operator
Key differentiator (Competitor advantage)	Dedicated vessel hardware; often subscription, licensing or training	carry licensing, cost, and training burdens, and the binding international conventions apply only to vessels 24 m and above, (we are targeting 9m boats) leaving the bulk of the global fleet outside the standard entirely	Each user must buy, carry and configure a dedicated device	Co-op deploys the system; fisher just connects with the phone they already carry; zero prior purchase, zero installation, and zero technical literacy Advantage:<SGD 50 Hardware 1.1M-Vessel Market 20–30 Boat Pilot Network Gets Stronger With Scale
Barriers to entry	Marine certification, regulatory approvals, satellite/carrier relationships, installed hardware and established service networks are expensive and slow to reproduce.	LoRa hardware and much of the research are publicly available; their advantage is mainly technical know-how and field-test data, not distribution or an established business network	Meshtastic's open ecosystem makes the core technology reproducible; established products gain protection through their hardware/software ecosystem, user base, brand and distribution.	Low technical, stronger operational moat over time. Commodity radios are easy to copy, but co-op relationships, regulatory approvals, installed gateways and accumulated fleet data become increasingly difficult to replicate.

VALIDATION

Can off- grid mesh connect to the internet?
Cruz et al. / C-Mesh (2025)



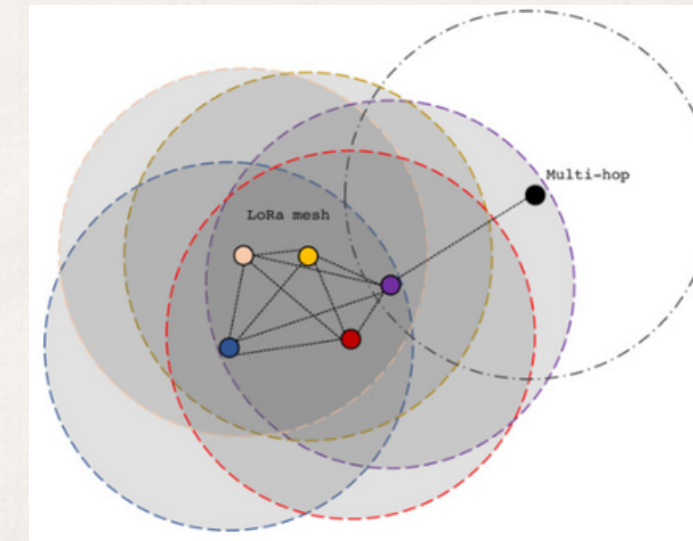
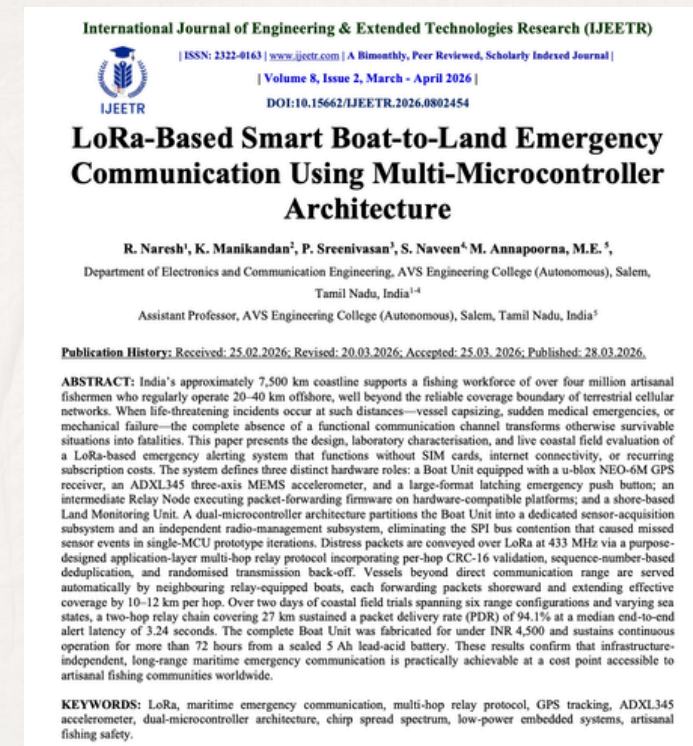
- C-Mesh could deliver data locally even when the land network server was unavailable.



Can off- grid mesh work in maritime applications?
Naresh et al. (2026)

Field-tested a LoRa communication system for artisanal fishermen on the Tamil Nadu coast.

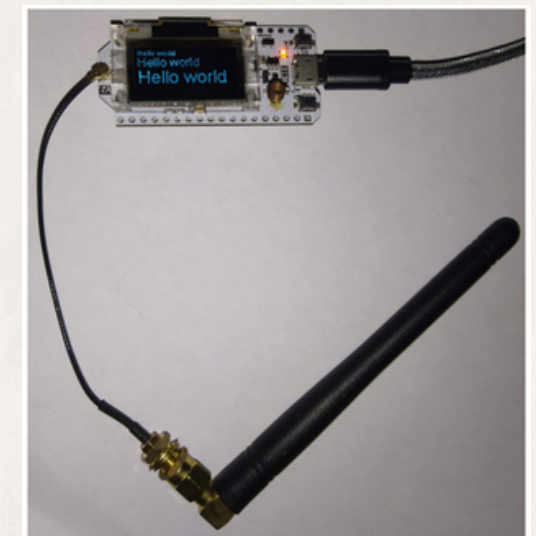
- Findings:-
- Field-reported range over water is 10–12 km, at 94% packet delivery.
 - <₹4,500 (US\$48) for their boat unit.
 - Boat-to-boat relaying; maritime propagation
 - Artisanal fishing as a real use case



Can off- grid mesh work?
Berto, Napoletano & Savi (2021)

The paper developed a gateway-free peer-to-peer LoRa mesh.

- Findings:-
- Work included inexpensive ESP32 LoRa hardware, around US\$20 in their reported prototype.
 - NODE A → NODE B → NODE C no internet



Milestones and usage of the awarded funding

PHASE 1 · MONTHS 1–6

Build & Prove It Works

- Design the message-relay protocol to fit **IMDA's 920-925MHz, 100mW, 1% duty-cycle limit**
- Build hardware: 15 boat units, 2 buoys, 1 shore gateway
- Test signal range & delivery rate in Singapore waters (**target: 90%+ delivered within 5km line-of-sight**)
- Simulate delivery reliability with only **20–30 boats spread** across a fishing ground
- Refine the **crew's no-app webpage (MVP built)** and build the co-op's shore-side dashboard
- Submit radio license application to IMDA (Singapore's telecom regulator)
- Start talks with fishing cooperations and marine insurers for the pilot

PHASE 2 · MONTHS 7–12

Pilot & Traction

- Install **20–30 boat units + 1 shore** gateway with a real fishing co-op
- Track message **delivery rate, speed, and system uptime**
- Write up pilot results as a **case study for investors & insurers**
- Get an insurer to commit to a data-sharing trial

BUDGET · \$10,000 SGD

- **Boat/buoy/gateway hardware** (17 units): \$3,500
- Range testing + IMDA application: \$1,300
- **Dashboard development**: \$400
- Travel + pilot install in **Indonesia/Philippines**: \$3,500
- **Case study** + next funding round prep: \$700
- Contingency: \$600
- Total: \$10,000